Benoit Clennett-Sirois

Principal Software Engineer @ Nasdaq
Building production AI agent platforms (FastAPI, Python, Asyncio, GCP)

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SUMMARY

I have been building software since 2009. Even when holding leadership titles, I stay a hands-on developer at heart and write code every day. My focus has always been on backend systems that are robust, secure, and thoughtfully architected.

Most recently at Nasdaq, I spearheaded the architecture and development of our business unit's AI agent platform. My focus was strictly technical leadership, building production systems with Python, FastAPI, asyncio, and LLM tooling like Agno and Vertex AI. One notable piece of work was building an automated optimization pipeline using evolutionary algorithms and synthetic datasets to tune agent team performance, prompts, and tool topologies.

Core Stack and Focus: Python, FastAPI, asyncio, Rust, GCP, Vertex AI, RAG engines, and distributed backend architectures.

EXPERIENCE

Nasdaq / Metrio

Permanent Full-time · 5 yrs

Principal Software Engineer (March 2025 - Present)

- Spearheaded the business unit's AI agentic platform development since November 2025.
- Agentic-based functionality development.
- Utilizing Agno, RAG Engines, and Vertex AI / Google Agent Platform.
- Backend stack: GCP, SQL, Cloud Run, FastAPI, and Pydantic.
- Automated agent optimization (including tools, prompts, and team topology) using an evolutionary algorithm via a synthetic dataset to rank agent team performance.

Lead Software Engineer (January 2023 - March 2025)

Software Developer at Metrio (January 2022 - February 2023) *Gatineau, Quebec, Canada*

Brûlerie Élixir

Founder (September 2020 - February 2023) *Gatineau, Quebec, Canada*

Shopify

4 yrs 7 mos

Development Manager (February 2020 - September 2020) *Greater Ottawa Metropolitan Area*

Senior Software Developer (March 2016 - February 2020) *Ottawa*

VarageSale

Web Developer (June 2014 - March 2016) *Greater Toronto Area, Canada*

- Web application developer, back-end and front-end.
- Technology stack includes Ruby on Rails, MySQL, Redis, and Memcached.
- Infrastructure maintenance and monitoring.

Caravan

Founding Member (April 2012 - June 2014) *Montreal*

- Employee-owned cooperative of long-time web designers and developers focused on sustainable client partnerships.

Tuktu.ca

Developer, Founder (March 2012 - June 2014) *Greater Montreal Metropolitan Area*

- Scalable web application for distributing and displaying classified ads.
- Technologies used: Python, Pyramid, MongoDB, PIL, Google V8 engine, jQuery, and dust.js.

Prana Biovegan Products

Web Developer (March 2009 - 2012) *Greater Montreal Metropolitan Area*

- Built the initial e-commerce application for Prana.
- Developed open-source payment and tax integration packages (`PyBeanstream`, `SatchmoBeanstream`, and `satchmo-canada-tax`).

Le site

Python Developer (May 2010 - December 2011) *Montreal*

- Worked on e-commerce and content sites using Django.
- Set up and managed multi-server architecture using Fabric for automatic deployments.

Yellow Pages Group / Groupe Pages Jaunes - Canada

Electronic Services (2008 - 2009)

- Data entry for Yellow Pages customer online profiles.

EDUCATION

- **École Internationale de l'Outaouais**
- **École secondaire Grande-Rivière** — High School Diploma (1995–2000)

LANGUAGES

- **English:** Native or Bilingual
- **French:** Native or Bilingual
- **Chinese:** Limited Working